

January 2026 CSE 220
Online on DFT and FFT
Fourier Hybrid Image
Subsections: A1, A2
Total Marks: 10
Total Time: 40 Minutes

September 5, 2026

The DFT separates broad, slowly varying image content from edges and fine details. A *hybrid image* takes these parts from different images: it resembles the high-frequency source from nearby and the low-frequency source from a distance.

You are given two existing offline images and a Gaussian kernel in `hybrid_image_template.py`. You must complete only the three blocks marked “TODO”. Do not modify `transforms.py`, `image_conv.py`, or any other offline file.

Mathematical model. Let the padded 2D spectra of the low-frequency source, high-frequency source, and Gaussian kernel be L , H , and G . If Δ is the spectrum of an impulse aligned with the kernel centre, construct

$$Y = LG + H(\Delta - G).$$

Here LG supplies the smooth component and $H(\Delta - G)$ supplies the detail component. Use one inverse transform and the usual linear-convolution crop.

An $R \times C$ image and a $K_r \times K_c$ kernel require at least

$$(R + K_r - 1) \times (C + K_c - 1)$$

transform samples. A radix-2 FFT requires each dimension to be increased to the next power of two.

Tasks

1. Complete `choose_transform_shape`. Determine the minimum linear-convolution dimensions and return dimensions supported by the selected transform engine. [2 Marks]
2. Complete `hybrid_plane`. Use the already prepared padded spectra to construct Y , perform exactly one inverse 2D transform, keep the real component, and return the correctly cropped plane. [5 Marks]
3. Complete `hybrid_image`. Process one plane for grayscale input or the three corresponding planes independently for RGB input, preserving the original image shape. [3 Marks]

Running and verification. First run the unmodified template and observe the indicated incomplete TODO. After completing all three TODOs, run:

```
python3 hybrid_image_template.py \  
  --low-image images/sunset512.png \  
  --high-image images/skyline512.png \  
  --kernel-size 31 --engine fft \  
  --out-dir outputs/lab_hybrid_student
```

The runner uses the sunset for low frequencies and the skyline for high frequencies. It writes the result, its components, a comparison figure, and a report. The report must show `verification: MATCH`; an error above 10^{-9} indicates an incorrect implementation.

Use only NumPy array arithmetic and functions imported by the template; `numpy.fft`, SciPy transforms, and library convolution routines are prohibited. Both grayscale and RGB inputs must work.